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THE RAJMAHAL TRAPS OF INDIA – K-Ar AGES AND PALAEOMAGNETISM

Ian McDOUGALL and M.W. McELHINNY
Department of Geophysics and Geochemistry, Australian National University, Canberra, A.C.T., Australia

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A number of concordant K-Ar ages in the range 100 to 105 my have been measured on samples from the tholeiitic Rajmahal Traps of India. These results are interpreted as giving a good estimate of the time of eruption of the basalts. By comparison with the physical time scale, this age is late Early Cretaceous (Albian). The isotopic ages can be reconciled with the conflicting views expressed by previous workers on the age of the Rajmahal Traps on the basis of palaeontological studies. Palaeomagnetic measurements on the basalts yield a palaeomagnetic pole in excellent agreement with previous work. The polar-wander path for the Phanerozoic of India is reviewed in the light of the age of the Rajmahal Traps and apparent inconsistencies in some of the previous data. This results in a reconsideration of the stratigraphic position of some of the beds sampled for palaeomagnetic purposes.

1. Introduction

Intrusions and lava flows of tholeiitic composition are very widespread in the late Palaeozoic to Mesozoic flat-lying sedimentary sequences in South Africa, South America, Tasmania, Antarctica and India. It has commonly been suggested that these are all synchronous and are related to a stage of the break-up of the great supercontinent of Gondwanaland. K-Ar dates are now available from all these provinces except India, where the Rajmahal Traps, a sequence of tholeiitic basalts, crop out in the Rajmahal Hills of West Bengal and Bihar. The age of these lavas has been the subject of considerable debate over many years; estimates of their age range from Late Triassic to Early Cretaceous. In an attempt to resolve this question, K-Ar ages have been measured on a number of samples from the Rajmahal Traps concurrently with palaeomagnetic measurements.

2. Geological setting and stratigraphic age

A voluminous literature exists on the Gondwana System of India, and summaries have been given by Wadia [1], Pascoe [2] and Krishnan [3]. The Gondwana System (or Group) consists of up to 6600 m of sedimentary rocks with relatively minor developments of volcanic rocks. The sediments are predominantly fluviatile and lacustrine, deposited in a number of intracratonic basins, and are now preserved in downfaulted blocks, mainly in northeast India. The Gondwana beds generally are undeformed and unmetamorphosed, and from palaeontological evidence were deposited from the Late Carboniferous to possibly the middle of the Cretaceous. Commonly the Gondwana System is divided into a Lower and an Upper Gondwana sequence.

In the Rajmahal Hills in West Bengal and Bihar, to the south and southwest of the River Ganges plain, the Rajmahal Series crops out over more than 3900 km^2. Ball [4] produced a geological map and description of the region. The Rajmahal Group is included within the Upper Gondwanas and comprises the Rajmahal Traps, a sequence of tholeiitic lavas at least 600 m thick, and intercalated sedimentary rocks, rarely more than 35 m in total thickness. At least 10 lava flows have been identified [5], and individual flows range from 20 m to 75 m thick. The Rajmahal Traps are essentially horizontal and rest more or less conformably, but with evidence of an erosional break, upon the Dubrajpur Stage of the Mahadeva

Series, or they lap onto older Gondwana sediments or the metamorphic basement. The Dubrajpur Stage consists of conglomerates and sandstones which contain but few fossils, mainly plants. Other stages of the Mahadeva Series from elsewhere in India, regarded as correlatives of the Dubrajpur, have yielded fish and reptile remains regarded as indicative of a Late Triassic age.

Some of the thin shale beds interbedded with the Rajmahal Traps contain an abundance of fossil plants, but there has been a great deal of controversy as to what age may be inferred from them. Age assignments include Late Triassic [7], Early Jurassic [6], Early to Late Jurassic [8], Middle to Late Jurassic [4,9,11] and Early Cretaceous [10]. Sah and Jain [12] from their studies of spores and pollen grains recognized that certain elements of the flora have an Early Cretaceous aspect but regarded the assemblage as transitional between Early Jurassic and Early Cretaceous. Biswas [13] also commented on the Early Cretaceous affinities of the floras.

In the Rajmahal Hills no younger formations lie upon the Rajmahal Group except for laterite and alluvium. However, to the east in the Bengal Basin, petroleum exploration bore holes have penetrated a sequence of Cainozoic and Cretaceous sedimentary rocks lying above tholeiitic basalts that are in apparent physical continuity with those of the Rajmahal Group [13]. These correlatives of the Rajmahal Traps are overlain in the bore holes by the Bolpur Formation with an erosional break between them, and Biswas [14] reports that the primitive dicotyledonous pollens found in the Bolpur Formation indicate an Early Cretaceous to early Late Cretaceous age. Thus, from the most recent palaeontological evidence it appears that the Rajmahal Group must be older than Late Cretaceous and probably in the range Late Jurassic to Early Cretaceous.

The Rajmahal Traps available for this study are tholeiitic basalts and were obtained from two areas, one near Rampurhat in West Bengal (24°11′N, 87° E) and the other from Mirza Chauki (25°15′N, 87° E) and Maharajpur (25°12′N, 87°45′E) in Bih... The basalts consist mainly of plagioclase and the... nopyroxenes pigeonite and augite, which are ge... lly free of alteration. A small proportion of pl... clase phenocrysts occurs in some samples, and ra... ystals of altered olivine were noted. All the samples contain from 3 to 10 per cent of a green, brown or reddish brown mineraloid, probably a form of chlorophaeite and/or palagonite. In some rocks this material appears to be an amygdaloidal infilling, but in others it occurs in irregular areas. Up to 5 per cent of pale brown glass is present in the basalts, usually slightly anisotropic, suggesting that some devitrification has taken place. All the rocks are even grained and intergranular with grain size 0.1 to 0.3 mm.

3. Potassium-argon age measurements

3.1. *Methods*

The techniques of K-Ar measurement were described previously [15,16]; argon was determined by isotope dilution and potassium by flame photometry. The error assigned to individual ages (table 1) is one standard deviation, calculated from the uncertainties in the physical measurements associated with each age determination, and averages about 2 per cent.

Whole rock samples were initially used for the measurements, despite the fact that none of the rocks was considered ideal for whole rock determination because of the presence of mineraloids and slightly anisotropic glass. The discordances between the whole rock ages made it imperative that minerals be separated. This was accomplished for a few samples, despite the fine grain size of the basalts.

3.2. *Results and discussion*

Data are listed in table 1. A point of interest is that all the whole rock samples have potassium contents in the range 0.24 to 0.44 per cent, considerably lower than values previously reported for Rajmahal basalts. The rocks are apparently low potassium tholeiites.

Only a single lava flow was sampled in the Rampurhat area, West Bengal, and hence two specimens were dated from it as a check on internal consistency. The two whole rocks and a plagioclase separated from one of these all give concordant ages at 100 ± 2 my. Both rocks contain about 10 per cent of mineraloid, but no glass was identified. The age is a reliable minimum, and the excellent agreement found suggests that the age probably dates the time of crystallization of the lava flow.

Samples from three basalt flows exposed in the

Table 1
Potassium-argon ages on Rajmahal Traps, India.

Sample No.	Field No.	Sample	K (wt %)	Rad. ^{40}Ar (10^{-11} mol/g)	$\frac{^{40}Ar_{rad}}{^{40}Ar}$ (%)	Calculated Age (my)	Locality	
GA 3200	RH 4	Whole rock	0.187, 0.187	3.38	81.9	99.0 ± 1.3	Same lava near Rampurhat, West Bengal	
GA 3200	RH 4	Plagioclase	0.110, 0.110	1.95	46.6	96.7 ± 3.6		
				2.03	77.7	100.5 ± 1.3		
GA 3201	RH 6	Whole rock	0.265, 0.265	4.95	52.9	102.1 ± 1.4		
GA 3202	RH 10	Whole rock	0.444, 0.444	6.83	43.6	84.6 ± 1.2	Mirza Chauki fourth flow	Bihar
				6.66	69.1	82.5 ± 1.1		
GA 3203	RH 11	Whole rock	0.254, 0.255	4.92	80.7	105.7 ± 1.4		
GA 3203	RH 11	Plagioclase	0.080, 0.082	1.55	63.6	104.2 ± 2.2	Second flow from base, Maharajpur	
				1.59	37.4	106.7 ± 2.3		
GA 3513	RH 14	Plagioclase	0.105, 0.106	1.90	58.7	98.4 ± 1.8		
				2.11	47.7	109.1 ± 2.1		
GA 3204	RH 15	Whole rock	0.240, 0.242	3.58	52.5	81.8 ± 1.2		
GA 3204	RH 15	Plagioclase	0.079, 0.080	1.47	60.7	101.0 ± 1.5	Lowest flow, Maharajpur	
				1.55	26.0	106.2 ± 2.1		
GA 3514	RH 16	Plagioclase	0.147, 0.149	2.41	58.7	88.1 ± 1.8		
				2.47	46.1	91.4 ± 1.4		

$\lambda_e = 0.585 \times 10^{-10}\ yr^{-1}$, $\lambda_\beta = 4.72 \times 10^{-10}\ yr^{-1}$, $^{40}K/K = 1.19 \times 10^{-2}$ atom %.

Bihar region were available for dating. The K-Ar ages on the whole rock samples form a complex pattern ranging from 82 to 106 my and cannot be reconciled with the known stratigraphy. The measured age of 106 my for the second flow from the base of the section (GA 3203) is similar to that found for the basalt from the Rampurhat area. It is apparent that the whole rock samples GA 3203 and GA 3204 have lost radiogenic argon relative to sample GA 3203. There is no obvious petrographic difference between the three samples as all contain a few per cent of fresh, but slightly anisotropic glass and several per cent of greenish brown or yellow mineraloid. Because of the discrepancies an attempt was made to separate plagioclase from each of the rocks. It proved too difficult in the case of sample GA 3202, but plagioclase was obtained from several other specimens. The plagioclase concentrate from GA 3514 is not pure owing to composite grains with glass.

Good concordancy is found between the ages on a whole rock and two plagioclase separates from the second flow at 104.5 ± 2 my. A virtually identical age was determined on a plagioclase from the lowest flow (GA 3204), but the impure plagioclase from another sample from this flow (GA 3514) has lost some radiogenic argon. The agreement of ages from two of the Maharajpur lavas and that from Rampurhat at 100 to 105 my is regarded as good evidence that this approximates closely to the time of eruption. It has been shown that loss of radiogenic argon has occurred from some of the Maharajpur rocks so that it is not possible to state unequivocally that no such loss has taken place from those samples that yield concordant ages. However, we can be confident that 100 to 105 my is a reliable minimum age for the outpouring of these lavas.

In a comprehensive review of data on the physical time scale, Casey [7] places the age of the boundary between the Early and Late Cretaceous at 100 my. Hence the results on the Rajmahal Traps suggest that they were erupted in the later part of the Early Cretaceous, ie, in Albian times. This assignment to the geological time scale is somewhat younger than that previously suggested for the Rajmahal Traps, but is

just consistent with the stratigraphic control provided by the microflora in the Bolpur Formation overlying the Rajmahal Group in the Bengal Basin [14]. The suggested age cannot be completely reconciled with the palaeontological estimates of age from the intertrappean beds, discussed earlier, although it is worth noting that with the progress of time and increasing knowledge of the local fossil floras, as well as those outside India, the age assigned to the Rajmahal flora has tended to become younger. At present the possibility that the Rajmahal flora is Early Cretaceous is admitted.

In summary the isotopic dating suggests a late Early Cretaceous age for the eruption of the Rajmahal Traps, and the palaeontological estimates of age extend up into the Early Cretaceous. The interpretation of all the available data seemingly provides a strong case for an Early Cretaceous, rather than a Late Jurassic or earlier age, for the outpouring of the Rajmahal Traps.

Samples have now been dated from each of the extensive developments of tholeiitic basaltic rocks that intrude or overlie late Palaeozoic to Mesozoic sedimentary sequences in the southern continents. The Karroo Dolerites of South Africa were emplaced over a relatively long time span from 150 to 190 my ago [18]. The basalts of South West Africa yield ages that concentrate at about 130 my [19], matching closely the main period of eruption of the very widespread Serra Geral basalts of South America [20, 21]. The Tasmanian and Antarctic tholeiites were emplaced at virtually the same time some 155-165 my ago [22, 18]. Hence the Rajmahal Traps, dated at about 100-105 my, were erupted at a considerably later time than the tholeiites of the other continents, which together with India formed Gondwanaland. The earlier idea that the eruptions in all the continents were synchronous is no longer tenable.

4. Palaeomagnetism

4.1. *Sampling and measurement*

Sixteen oriented hand samples were collected from eight different horizons in the traps. The K-Ar measurements were made on some of these samples. Although only five flows are represented, some of them are so thick (≈ 70 m) that several horizons were sampled as separate points in time.

From each oriented block cylinders 2.2 cm in diameter and 2.2 cm long were cored and sliced in the laboratory, and their magnetizations measured on astatic magnetometers. NRM directions were tightly grouped and the samples exhibited excellent behaviour during alternating field demagnetization using an apparatus similar to that described by McElhinny [23]. Very small secondary components were removed in most cases and the original tight grouping of the site mean directions improved in precision from 69 to 187 after

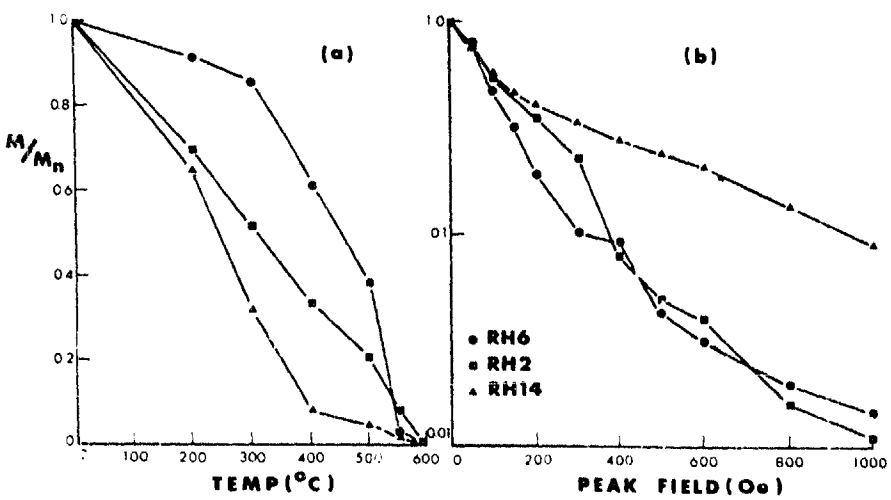


Fig. 1. (a) Thermal, (b) Alternating field demagnetization curves for samples from the Rajmahal Traps.

Table 2
Palaeomagnetic results from the Rajmahal Traps, India, after magnetic cleaning in 200 Oe peak field.

Locality	Samples	D	I
Barapahari quarries nr. Rampurhat, W. Bengal (24°11'N, 87°44'E)	RH 1, 2	309	−70
	RH 3, 4	302	−73
	RH 5, 6	320	−65
Banshoi River nr. Amrapara, Bihar (24°30'N, 87°35'E)	RH 7, 8	296	−70
Murali quarry, 4th flow nr. Mirza Chauki, Bihar (25°15'N, 87°30'E)	RH 9, 10	315	−71
Maharajpur (25°12'N, 87°45'E) 2nd flow from base	RH 11, 12	311	−62
	RH 13, 14	316	−62
1st flow	RH 15, 16	305	−60
MEAN (24.8 N, 87.7 E)	N=8 sites	310 (k = 187, α_{95} = 4)	−67
POLE POSITION	N=8	Lat. 3S, Long. 118E (K = 91, A_{95} = 6)	

magnetic cleaning in 200 oersteds peak field. The samples were extremely stable even in fields up to 800 or 1000 oersteds peak value.

During thermal demagnetization, which was carried out using the apparatus described by Irving et al. [24], the samples were stable up to the Curie Point of magnetite, any small secondary components being removed at temperatures of 200 or 300°C. Alternating field and thermal demagnetization curves are compared in fig. 1. The a.f. curves are typical of stable basalts, but the thermal curves decay much more quickly than expected at the lower temperatures. In spite of this, the rocks are obviously very stable magnetically and there is little doubt that their magnetization relates to the time of extrusion.

4.2. *Results*

Two specimens from each sample were magnetically cleaned in 200 oersteds peak field. The site mean directions are given in table 2, together with the overall formation direction and palaeomagnetic pole position. This can be compared with the result obtained by Radhakrishnamurthy [25] from a more detailed study of the traps at fifteen sites (table 3). The agreement between the two investigations made in different laboratories is gratifying and serves as an excellent interlaboratory comparison.

5. The Phanerozoic polar-wander curve for India

A number of new palaeomagnetic results have recently been published for the Phanerozoic of the Indian subcontinent. These alter the previous simple picture of the polar wander path for India [26]. A complete list of Phanerozoic results is given in table 3. Only those results derived from magnetic or thermal cleaning studies have been listed, so that some of the early work reported for India [27,28] has been omitted because these results are now superseded by more comprehensive studies. A polar-wander path for the Phanerozoic of India has been drawn in fig. 2. There are several important points to be noted. Between the Early Cambrian and Late Carboniferous no data are available; therefore these two points have been joined by a dashed line along the great circle through them. The angular distance is about 110°, whereas it is normal practice in drawing polar-wander

Table 3
Phanerozoic palaeomagnetic results from India.

No.	Formation	Age	Pole position		A_{95}	Reference
1	Purple sanstone, Salt Range	L. Cambrian	28S	32E	11°	[34]
2	Talchir Series	U. Carboniferous	32N	134E	3°	[35]
3	Kamthi Beds (Wardha Valley)	U. Permian	21N	130E	10°*	[30]
4	Mangli Beds	L. Triassic	7N	124E	6°	[30]
5	Kamthi (Panchet?) Beds (Godavary Valley)	L. Triassic**	19S	128E	7°*	[31]
6	Parsora Sandstones	U. Triassic	30S	125E	6°*	[33]
7	Pachmarhi Beds	U. Triassic	10S	130E	10°*	[30]
8	Rajmahal Traps	L. Cretaceous	12S	114E	5°	[25]
9	Rajmahal Traps	(Albian 100 my)	3S	118E	6°	This paper
10	Sylhet Traps	L. Cretaceous?	16S	120E	8°	[36]
11	Tirupati Sandstones	U. Cretaceous**	30S	107E	5°	[32]
12	Lower Deccan Traps	U. Cretaceous to	30S	101E	3°	[37,38]
13	Upper Deccan Traps	Palaeocene [39]	37S	101E	8°	[37,38]

* Recalculated from original data giving unit weight to each site.
** Small changes in age are proposed for these sediments as explained in the text

paths to join successive poles through the shortest path, bearing in mind that both polarities are possible. However, a comparison of the Palaeozoic results from India with those from the other Gondwana continents [29] suggests that this path is the most appropriate.

There are two results from the Kamthi Beds (Nos. 3 and 5) from widely separated localities which do not agree. Wensink [30] has noted this apparent inconsistency and suggested that this might be due to the incompleteness of the demagnetization procedures. The more likely explanation can be found in the difficulty of correlating the Kamthi Beds down the Wardha-Godavary Valley [2,3] where they become indistinguishable from the Lower Triassic Panchet Series. Verma and Bhalla [31] admit the possibility of an Early Triassic (Panchet) age for their sampling sites (pole 5) and only tentatively assigned a Late Permian (Kamthi) age to them. The Mangli Beds (pole 4) of the Wardha Valley are also more or less similar in appearance to the Kamthi Beds and referable to the Panchet Series on fossil evidence [3].

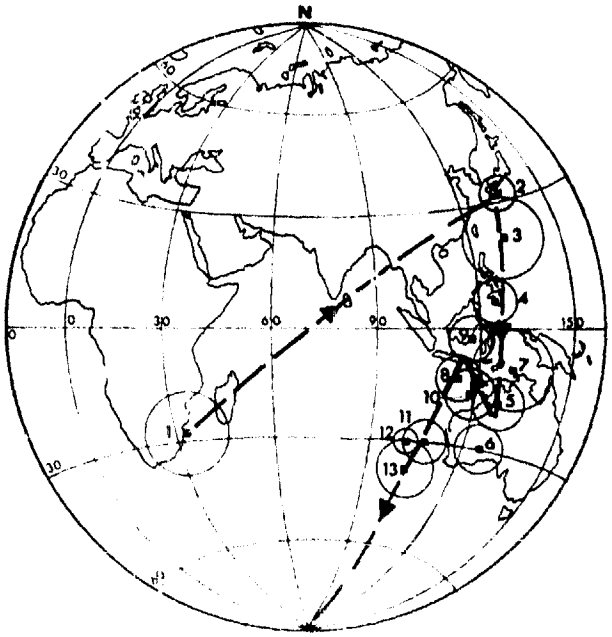


Fig. 2. Phanerozoic polar-wander path for India. Poles are numbered as in table 3 and circles of 95% confidence are shown. Equal area meridional projection.

We have there therefore preferred the interpretation given in table 3, and, in making a minor modification to the age of pole 5 suppose that these Kamthi (Panchet?) Beds are stratigraphically slightly younger than the Mangli Beds of pole 4, both being Lower Triassic. With this change a logical and continuous polar-wander path results as illustrated in fig. 2. It is worth noting that the Kamthis of the Wardha Valley (pole 3) are all reversely magnetized and they could therefore lie within the Kiaman Magnetic Interval, consistent with their designated Late Permian age, whilst those of the Godavary Valley (pole 5) are all normally magnetized and therefore cannot lie within the Kiaman interval.

There is a 40° difference between the late Palaeozoic poles [2,3] and the group of Triassic poles [5–7] with pole 4 apparently occupying a position intermediate between the two groups. The polar movement between these groups must have occurred fairly rapidly during the Early Triassic if the ages assigned in table 3 are accepted. Assuming, as is usual [3], that the Sylhet Traps are the same age as the Rajmahal Traps, the three mid-Cretaceous poles [8–10] all lie to the west of the Triassic group [5–7]. It appears that there was a short westerly movement of the pole (as shown in fig. 2) between the Late Triassic and mid-Cretaceous.

The Tirupati Sandstones (pole 11) have been cited as "Middle" Cretaceous [32], but in the Godavary Valley, they overlie sandstones containing flora allied to the Rajmahal Stage [3]. The revised age of the Rajmahal Traps therefore suggests that these beds are Late Cretaceous in age, and they have been so designated in table 3. In support of this, we note that there is a significant difference between the three mid-Cretaceous poles and the Tirupati pole, which itself lies close to that for the Lower Deccan Traps, whose age has recently been determined as Late Cretaceous to Palaeocene around 65 my [39]. There is therefore a southerly movement of the pole during the Late Cretaceous of about 20°, with the final 50° of movement to its present position taking place during the Tertiary after the extrusion of the Deccan Traps [38].

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